Kubelet Resolution Comparison Report

Generated 2026-05-31T14:45:44.277Z | Scope: post-remediation kubelet analysis for ensemble-grafana

Bottom line: The inventory-service probe tuning is deployed and the short post-change window shows `0` probe failures. Remaining kubelet findings are containerd/kubelet `NotFound` cleanup messages on `ip-10-42-101-208`, which should be monitored for recurrence.

Implemented Change

Probe	Before	After
Readiness	initialDelay=60s, period=10s, default timeout/failure threshold	initialDelay=60s, period=10s, timeout=5s, failureThreshold=5
Liveness	initialDelay=90s, period=20s, default timeout/failure threshold	initialDelay=120s, period=20s, timeout=5s, failureThreshold=5

Query Results

Six-Hour Error/Failure Scan

18 matching kubelet log samples across 5 streams.

Category	Count
Probe failed	10
ContainerStatus NotFound	4
DeleteContainer NotFound	4

Post-Change Window

4 matching samples.

Signal	Count
Probe failed	0
Runtime/container NotFound	4

Probe Failure Grouping (Six-Hour Window)

Service	Pod	Probe	Node	Count
inventory-service	ensemble-grafana/inventory-service-77c95f9b47-2mfc8	Readiness	ip-10-42-101-134.ec2.internal	1
inventory-service	ensemble-grafana/inventory-service-77c95f9b47-cbft7	Liveness	ip-10-42-101-208.ec2.internal	2
inventory-service	ensemble-grafana/inventory-service-77c95f9b47-cbft7	Readiness	ip-10-42-101-208.ec2.internal	7

Codex Resolution Plan

- Implemented inventory-service probe tuning: readiness timeoutSeconds=5, failureThreshold=5; liveness initialDelaySeconds=120, timeoutSeconds=5, failureThreshold=5.
- Rollout completed successfully and the current inventory-service pod is Ready with 0 restarts.
- Post-change window has 0 kubelet probe failures; inventory-service probe failures are not recurring immediately after the rollout.
- Residual kubelet samples are 4 container runtime NotFound/DeleteContainer cleanup events on ip-10-42-101-208.
- Next action is monitoring rather than another app probe change: check whether containerd NotFound recurs during future rollouts/load tests and correlate with node/runtime logs.

Grafana Assistant Resolution Plan

I'll analyze the kubelet ERROR-level messages from your ensemble-grafana cluster to assess the current state after inventory-service probe tuning.

Resolution Plan

Current State: Probe tuning successful. No inventory-service probe failures in 3-hour window post-rollout.

Active Root Cause: Container lifecycle mismatch between kubelet and containerd runtime on node `ip-10-42-101-208.ec2.internal` at 10:44:06 EDT. Kubelet queried 2 containers already removed from containerd.

Resolution Steps:

1. **Verify probe tuning success ✓** — No liveness/readiness failures detected; inventory-service probe issue resolved

2. **Investigate containerd sync** — Check containerd logs on `ip-10-42-101-208` around 10:44:06 for premature garbage collection

3. **Review pod evictions** — Confirm if affected containers belonged to terminated pods; verify grace period settings

4. **Monitor pattern** — Track if container-not-found errors recur; isolated timestamp suggests transient event

Assessment: inventory-service probe failures **no longer appear** after tuning. Current issue is unrelated runtime cleanup timing on single node.

Side-by-Side Comparison

Area	Codex	Grafana Assistant	Readout
Probe tuning result	No post-change probe failures in the short window	Probe tuning successful; no inventory-service probe failures detected post-rollout	Aligned.
Current active signal	Runtime NotFound cleanup events on ip-10-42-101-208	Container lifecycle mismatch between kubelet and containerd on ip-10-42-101-208	Aligned.
Next action	Monitor recurrence and correlate with rollouts/load before node/runtime changes	Check containerd logs, review evictions, monitor pattern	Aligned; Codex recommends monitoring first due isolated burst.

Recommended Next Actions

- Keep the inventory-service probe tuning deployed.
- Run the next traffic/load test and watch the dashboard kubelet tab for fresh Probe failed series by service/pod/probe/node.
- If only containerd NotFound events recur, investigate node ip-10-42-101-208 containerd/kubelet logs and pod termination timing around rollout windows.
- Escalate to kubelet/containerd GC or node replacement only if NotFound bursts recur independently of normal pod deletion.

Post-Change Samples

Time	Node	Pod	Container	Probe	Output
2026-05-31T14:44:06.507Z	ip-10-42-101-208.ec2.internal				E0531 14:44:06.507703 2212 log.go:32] "ContainerStatus from runtime service failed" err="rpc error: code = NotFound desc = an error occurred when try to find container

Time	Node	Pod	Container	Probe	Output
					\"814c9548a6304d4d28c8f5adfeaec08d10ee32cc02a08885f034540edf5333f4\": not found" containerID="814c9548a6304d4d28c8f5adfeaec08d10ee32cc02a08885f034540edf5333f4"
2026-05-31T14:44:06.507Z	ip-10-42-101-208.ec2.internal				I0531 14:44:06.507743 2212 pod_container_deletor.go:53] "DeleteContainer returned error" containerID={"Type":"containerd","ID":"814c9548a6304d4d28c8f5adfeaec08d10ee32cc02a08885f034540edf5333f4"} err="failed to get container status \"814c9548a6304d4d28c8f5adfeaec08d10ee32cc02a08885f034540edf5333f4\": rpc error: code = NotFound desc = an error occurred when try to find container \"814c9548a6304d4d28c8f5adfeaec08d10ee32cc02a08885f034540edf5333f4\": not found"
2026-05-31T14:44:06.508Z	ip-10-42-101-208.ec2.internal				E0531 14:44:06.508319 2212 log.go:32] "ContainerStatus from runtime service failed" err="rpc error: code = NotFound desc = an error occurred when try to find container \"5e3f368171f47a43262b9a64abfc7e453856169aa43dfa409f862a314312e44c\": not found" containerID="5e3f368171f47a43262b9a64abfc7e453856169aa43dfa409f862a314312e44c"
2026-05-31T14:44:06.508Z	ip-10-42-101-208.ec2.internal				I0531 14:44:06.508345 2212 pod_container_deletor.go:53] "DeleteContainer returned error" containerID={"Type":"containerd","ID":"5e3f368171f47a43262b9a64abfc7e453856169aa43dfa409f862a314312e44c"} err="failed to get container status \"5e3f368171f47a43262b9a64abfc7e453856169aa43dfa409f862a314312e44c\": rpc error: code = NotFound desc = an error occurred when try to find container \"5e3f368171f47a43262b9a64abfc7e453856169aa43dfa409f862a314312e44c\": not found"

Commands

```
kubectl apply -f infra/k8s/services.yaml
kubectl -n ensemble-grafana rollout status deployment/inventory-service --timeout=180s

gcx logs query -d grafanacloud-logs '{job="integrations/kubernetes/journal", k8s_cluster_name="ensemble-grafana",
unit="kubelet.service"} |~ "(?i)error|warning|failed|evict|oom|backoff|crash"' --since 6h -o json --limit 100 --no-truncate

gcx logs query -d grafanacloud-logs '{job="integrations/kubernetes/journal", k8s_cluster_name="ensemble-grafana",
unit="kubelet.service"} |~ "(?i)error|warning|failed|evict|oom|backoff|crash"' --since 15m -o json --limit 100 --no-truncate

gcx assistant prompt --context assistant-oauth --timeout 300 'Use Grafana Cloud logs to analyze kubelet ERROR-level/error messages
for cluster ensemble-grafana after tuning inventory-service readiness/liveness probes. Query: {job="integrations/kubernetes/journal",
k8s_cluster_name="ensemble-grafana", unit="kubelet.service"} |~ "(?i)error|warning|failed|evict|oom|backoff|crash". Determine current
root causes and produce a concise Resolution Plan, noting whether inventory-service probe failures still appear after the probe-
tuning rollout.'
```